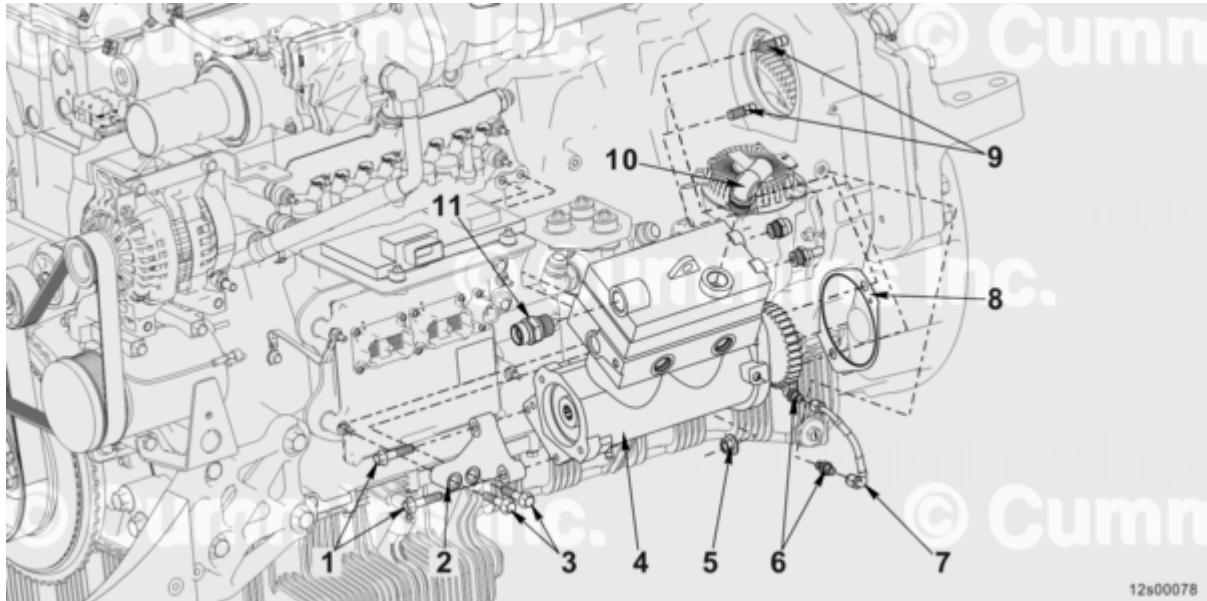


Trainings ▾

Exploded View



Air Compressor Exploded View

1. Capscrew - quantity 2
2. Brace - quantity 1
3. Capscrew - quantity 2
4. Air compressor - quantity 1
5. Nut - quantity 2
6. Connector - quantity 2
7. Oil hose - quantity 1
8. Gasket - quantity 1
9. Stud - quantity 2
10. Air inlet connector - quantity 1
11. Air outlet connector - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

- No additional service items required.

Pressure Test

WARNING

The external pressure tank used must meet SAE J10 and FMVSS121 standards, and have a safety pressure relief valve which opens between 1034 to 1207 kPa [150 to 175 psi]. Failure to use the proper pressure vessel and plumbing can result in property damage and serious personal injury.

1. Park the vehicle. Use wheel chocks or an appropriate anti-roll device to stabilize the vehicle.
2. Drain the vehicle air system.
3. Remove the air discharge hose and air governor signal hose from the air compressor. Install pipe plugs into the air compressor unloader signal ports.
4. Plumb the air compressor discharge hose into an external pressure tank. The external pressure tank **must** be equipped with a pressure gauge and pressure relief valve. Make sure that the fittings are installed with appropriate thread sealant and do **not** leak.

Start the engine and operate the engine at 2000 rpm.

Note : Do not cycle air accessories such as seats, doors, wipers, air bags, and so forth.

Once the external pressure tank pressure reaches 800 kPa [116 psi], shut the engine off. The maximum buildup time is 3 minutes.

Verify that the air compressor will build pressure in the external tank. If air pressure successfully builds to 800 kPa [116 psi] within the time limit, the air compressor functions properly. Remove the external air discharge hose from the air compressor. Install the vehicle air discharge hose and air governor signal hose to the compressor. See the Air Compressor Will **Not** Pump Air troubleshooting symptom tree in Section TS in 15N CM2380 M104B Fault Code Troubleshooting Manual, Bulletin 5659765, to inspect the rest of the air system components for leaks and proper operation.

If the air compressor fails to build to 800 kPa [116 psi] within the time limit, the air compressor is malfunctioning and the cylinder head needs to be repaired, based on the marking scribed on the head. Refer to Procedure 012-013 in Section 12.

(/qs3/pubsys2/xml/en/procedures/1016/1016-012-013.html)

Preparatory Steps

WARNING

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

WARNING

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Use steam to clean the air compressor. Dry with compressed air.
- Drain the engine coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the hydraulic pump drive. Refer to Procedure 009-016 in Section 9.
(/qs3/pubsys2/xml/en/procedures/1016/1016-009-016.html)
- Remove the air compressor inlet tube. Refer to Procedure 012-109 in Section 12.
(/qs3/pubsys2/xml/en/procedures/1016/1016-012-109.html)
- Remove the air compressor coolant lines. Refer to Procedure 012-004 in Section 12.
(/qs3/pubsys2/xml/en/procedures/1016/1016-012-004.html)
- Remove the Closed Crankcase Ventilation (CCV) hoses. Refer to Procedure 003-024 in Section 3. (/qs3/pubsys2/xml/en/procedures/1016/1016-003-024-tr.html)
- Remove the low-pressure fuel supply lines. Refer to Procedure 005-248 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-248.html)

Remove

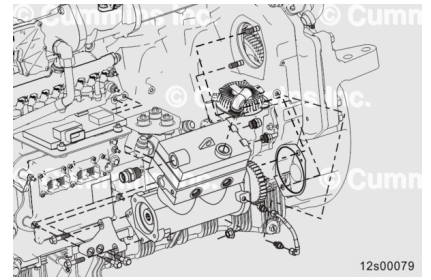
Remove the air compressor support bracket and capscrews.

Remove the oil hose.

Remove the two mounting nuts and the air compressor.

Discard the gasket.

Remove the studs and connectors if required.



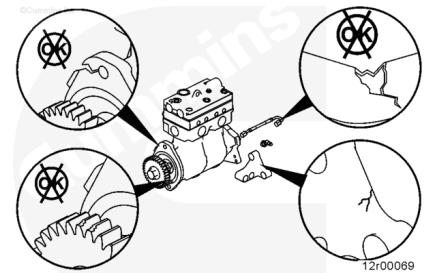
Clean and Inspect for Reuse

Inspect the compressor housing, hose, connector, and brace for cracks or other damage.

If cracks or other damage is found, replace the damaged component.

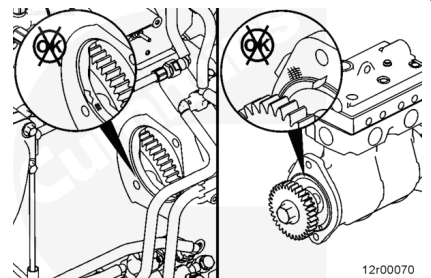
Inspect the compressor drive gear for cracks or other damage.

If cracks or other damage is found, replace the air compressor.



Make sure the gasket surfaces of the cylinder block and the air compressor are clean and **not** damaged.

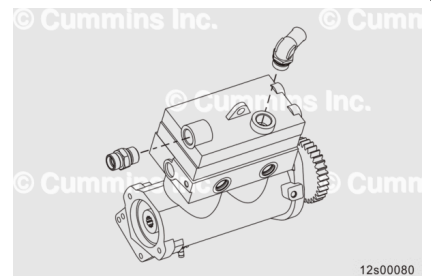
If damage is found on the gasket surface that will result in a leak path, the damaged component **must** be replaced.



Install

Tighten the air inlet connector or air outlet connector to air compressor, if removed.

Torque Value: 77 n•m [57 ft-lb]



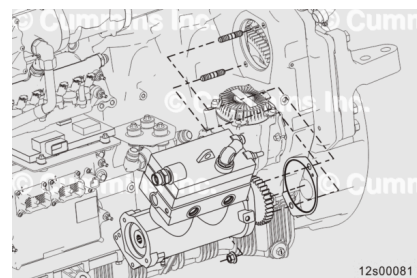
Tighten two studs into cylinder block, if removed.

Torque Value: 34 n•m [25 ft-lb]

Install the air compressor and the two nuts onto the studs.

Tighten the nuts.

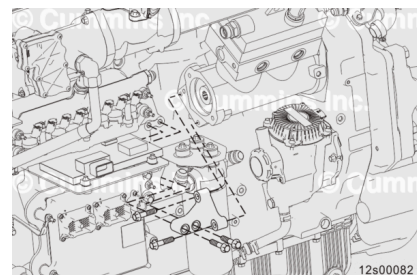
Torque Value: 75 n•m [55 ft-lb]



Install the air compressor mounting brace to the cylinder block and the air compressor.

Tighten the capscrews on the cylinder block and air compressor.

Torque Value: 47 n•m [35 ft-lb]

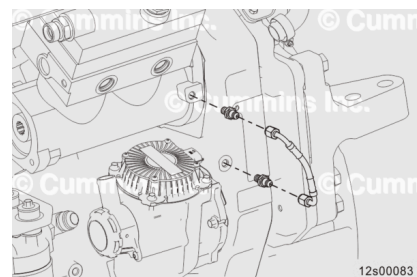


Tighten connectors to cylinder block and air compressor if removed.

Torque Value: 20 n•m [177 in-lb]

Install oil hose. Tighten two fittings.

Torque Value: 20 n•m [177 in-lb]



Finishing Steps

- Install the low-pressure fuel supply lines. Refer to Procedure 005-248 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-248.html)
- Install the CCV hoses. Refer to Procedure 003-024 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-024-tr.html)
- Install the coolant lines and fittings. Refer to Procedure 012-004 in Section 12.
(/qs3/pubsys2/xml/en/procedures/1016/1016-012-004.html)
- Install the air compressor inlet tube. Refer to Procedure 012-109 in Section 12.
(/qs3/pubsys2/xml/en/procedures/1016/1016-012-109.html)
- Install the hydraulic pump drive. Refer to Procedure 009-016 in Section 9.
(/qs3/pubsys2/xml/en/procedures/1016/1016-009-016.html)
- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Operate the engine to normal operating temperature and check for leaks.

